



Symbiosis Institute of Technology, Pune

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No.: 2

Name of Student	Akshit Gawade
PRN No.	25070122029
Batch	2025-2029
Class	A1
Academic Year & Semester	Second Year Semester III
Date of Submission	4/7/26
Title of Assignment:	Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.
Problem Statement Learning Objectives	Problem Statement Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management. Learning Objectives To understand and implement default, parameterized, and copy constructors for object initialization.
Theory	Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Program Code

```
1 class Students {
2     int id;
3     String name;
4
5     Students() {
6         this.id = 0;
7         this.name = "Unknown";
8     }
9
10    Students(int id, String name) {
11        this.id = id;
12        this.name = name;
13    }
14
15    Students(Students other) {
16        this.id = other.id;
17        this.name = other.name;
18    }
19
20    void display() {
21        System.out.println("Student ID: " + id + ", Name: " + name);
22    }
23
24    Run | Debug
25    public static void main(String[] args) {
26        Students s1 = new Students();
27        Students s2 = new Students(id: 2, name: "Alice");
28        Students s3 = new Students(s2);
29
30        s1.display();
31        s2.display();
32        s3.display();
33    }
```

Output Screenshots

```
Problems 4 Output Debug Console Terminal Ports
● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 2> & 'c:\Programing\Code\User\workspaceStorage\8ac64c362bf5428f8b6beb599cac9a72
Student ID: 0, Name: Unknown
Student ID: 2, Name: Alice
Student ID: 2, Name: Alice
○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 2>
```

Exercise Qns

1. Create a **Student** class using default and parameterized constructors to initialize student name and roll number.

```

Students.java > Students > main(String[])
1  class Students {
2      int id;
3      String name;
4
5      public static void main(String[] args) {
6          Students s1 = new Students();
7          s1.id = 1;
8          s1.name = "John Doe";
9          Students s2 = new Students();
10         s2.id = 2;
11         s2.name = "Alice";
12         System.out.println("Student ID: " + s1.id + ", Name: " + s1.name);
13         System.out.println("Student ID: " + s2.id + ", Name: " + s2.name);
14     }
15 }
16

```

2. Develop a **Mobile Phone Inventory System** using different constructors to initialize mobile details and create duplicate object records.

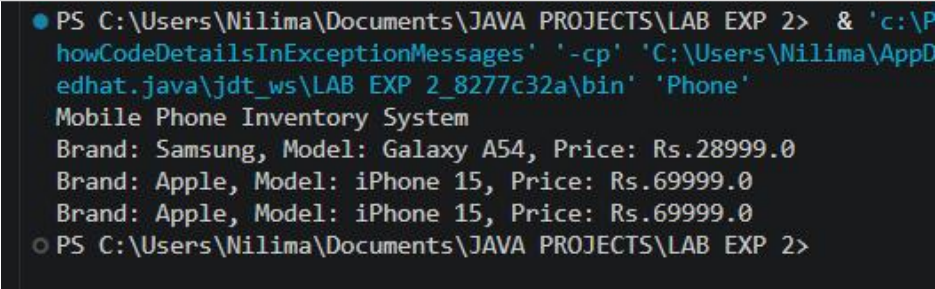
```

1  public class Phone {
2      String brand;
3      String model;
4      double price;
5
6      // Default constructor
7      Phone() {
8          brand = "Samsung";
9          model = "Galaxy A54";
10         price = 28999;
11     }
12
13     // Parameterized constructor
14     Phone(String brand, String model, double price) {
15         this.brand = brand;
16         this.model = model;
17         this.price = price;
18     }
19
20     // Copy constructor
21     Phone(Phone p) {
22         this.brand = p.brand;
23         this.model = p.model;
24         this.price = p.price;
25     }
26
27     void display() {
28         System.out.println("Brand: " + brand + ", Model: " + model + ", Price: Rs." + price);
29     }
30
31     Run | Debug
32     public static void main(String[] args) {
33         Phone phone1 = new Phone();
34         Phone phone2 = new Phone(brand: "Apple", model: "iPhone 15", price: 69999);
35         Phone phone3 = new Phone(phone2);
36
37         System.out.println(x: "Mobile Phone Inventory System");
38         phone1.display();
39         phone2.display();
40         phone3.display();
41     }
42 }

```

**Output
Screenshots**

Qn 1 completed in main output.
Qn 2:

	
Conclusion	This assignment demonstrated how default, parameterized, and copy constructors enable flexible object initialization and secure data management in Java. The exercises proved that custom copy constructors prevent memory reference conflicts, ensuring independent data replication. Ultimately, mastering these constructor methodologies reinforces core OOP principles, directly improving code reliability and maintainability.